MAMAMIND: AI-POWERED MATERNAL WELLNESS AND FAMILY SUPPORT SYSTEM

A PROJECT REPORT

Submitted by

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In fulfilment for the course

GE23627 – DESIGN THINKING AND INNOVATION

for the degree of

BACHELOR OF ENGINEERING
IN COMPUTER SCIENCE AND
DESIGN



RAJALAKSHMI ENGINEERING COLLEGE
THANDALAM, CHENNAI
OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

THANDALAM, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled **"MAMAMIND: AI-POWERED MATERNAL WELLNESS AND FAMILY SUPPORT SYSTEM WITH WEARABLE HEALTH MONITORING"** is the bonafide work of **GOKUL KRISHNA I S (241701016)** and **SRIYAANCHITA S (241701057)**, carried out under my supervision. Certified further that this work has not previously formed part of any thesis or dissertation submitted for a degree or award.

Certified further that the project contributes to **SDG 3, SDG 5 and SDG 9**, primarily focusing on **SDG 3 – Good Health and Well-Being**, through an AI-powered maternal wellness system integrating wearable health monitoring, personalized self-care, intelligent assistance and collaborative parenting to promote maternal well-being and family support during pregnancy and postpartum recovery.

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ABSTRACT

Maternal wellness and family support are essential during pregnancy, postpartum recovery and early childcare. However, mothers often face challenges in maintaining self-care routines, managing emotional well-being and coordinating baby-care responsibilities while receiving limited continuous support. MAMAMIND: AI-Powered Maternal Wellness and Family Support System with Wearable Health Monitoring addresses these challenges through an integrated mobile application and wearable monitoring band. The application is developed using React Native, Expo and TypeScript, with Firebase Authentication and Cloud Firestore supporting user accounts, activity records and shared family data. Dedicated Mother, Father and Baby dashboards provide personalized features for different caregiving roles. The Mother dashboard includes an AI voice companion, medication reminders, wellness check-ins, guided meditation, mood journaling, relaxation games and outing checklists. The Father dashboard offers daily support suggestions, shared tasks, encouragement messages, bonding activities and AI-assisted parenting guidance. The Baby dashboard supports feeding, sleep and diaper logs, vaccination reminders, milestones, memories and age-based food guidance. An ESP32-based wearable band monitors heart rate, blood oxygen saturation (SpO_2) and physical movement to support passive wellness awareness. When elevated heart rate persists, the system is designed to consider activity context and suggest appropriate rest or relaxation without presenting the readings as a medical diagnosis. Google Gemini integration enables conversational assistance and personalized suggestions, while Firestore supports coordination between family members. The project demonstrates a family-centered approach to combining artificial intelligence, wearable sensing and digital caregiving tools for maternal wellness support.

ACKNOWLEDGEMENT

Initially, we thank the Almighty for being with us through every walk of our lives and showering His blessings throughout the endeavour to complete this report. Our sincere thanks to our Chairman Mr. S. Meganathan, B.E., F.I.E., our Vice Chairman Mr. Abhay Shankar Meganathan, B.E., M.S., and our respected Chairperson Dr. (Mrs.) Thangam Meganathan, Ph.D., for providing us with the requisite infrastructure and facilities at their esteemed institution.

Our sincere thanks to Dr. S. N. Murugesan, M.E., Ph.D., our beloved Principal, for his kind support and the facilities provided to complete our project work successfully. We express our sincere thanks to Dr. S. Vinodh Kumar, Ph.D., Professor and Head of the Department of Computer Science and Design, for his valuable guidance and encouragement throughout the project. We convey our heartfelt gratitude to our internal project guide, Dr. P. Revathy, Professor, Department of Computer Science and Design, for her continuous guidance, constructive suggestions and valuable support during the development of our project. We also thank the faculty members, our families and friends for their encouragement and support.

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INTRODUCTION

1.1 General

Maternal health and well-being play an important role during pregnancy, postpartum recovery and early childcare. Mothers frequently experience physical changes, emotional challenges and increased responsibilities while caring for their newborns. Maintaining self-care routines, monitoring wellness and managing family responsibilities can become difficult without adequate assistance. Conventional approaches often depend on manual health tracking, separate wellness applications and limited family coordination.

MAMAMIND: AI-Powered Maternal Wellness and Family Support System with Wearable Health Monitoring addresses these challenges through an integrated mobile application and wearable monitoring system. The system combines AI-powered assistance, passive physiological monitoring, personalized self-care recommendations, baby-care management and collaborative parenting features. Dedicated Mother, Father and Baby dashboards provide a connected platform for supporting maternal wellness and family involvement.

1.2 Background

Digital healthcare and wellness applications have increasingly supported personal health tracking, medication management and emotional well-being. However, maternal wellness requires a more comprehensive approach that considers physical health, emotional support, childcare responsibilities and family participation. Many existing applications focus on individual activities rather than integrating maternal wellness with shared parenting and continuous monitoring.

Advancements in artificial intelligence, wearable sensors and mobile technologies enable more accessible and personalized wellness solutions. The ESP32 microcontroller supports sensor interfacing and wireless data transmission, while physiological sensors provide heart-rate, blood oxygen saturation and movement measurements. React Native enables cross-platform application development, while Firebase supports authentication and synchronized data storage. MamaMind integrates these technologies with AI-powered assistance to establish a connected maternal wellness and family-care platform

1.3 Problem Statement

Managing maternal wellness during pregnancy and postpartum recovery remains challenging due to physical changes, emotional demands, irregular routines and increased childcare responsibilities. Mothers may struggle to maintain medication schedules, monitor their well-being and allocate sufficient time for relaxation. Conventional wellness applications often require manual data entry and may not provide timely, personalized recommendations based on physiological observations.

Furthermore, maternal self-care, baby-care tracking and family coordination are frequently managed through separate applications or manual methods. This fragmented approach can reduce convenience and limit shared responsibility between parents. MamaMind addresses the need for an integrated system that combines wearable-based monitoring, AI-powered wellness assistance, personalized self-care, baby-care management and collaborative family support within a unified mobile application.

1.4 Motivation

The motivation behind MamaMind is to provide accessible and continuous wellness support for mothers during pregnancy and postpartum recovery. Mothers frequently prioritize newborn care and household responsibilities over their own physical and emotional well-being. Managing medication reminders, maintaining relaxation routines and communicating support needs can become difficult without an organized and supportive system.

MamaMind was developed to connect intelligent digital assistance with passive wearable monitoring and collaborative parenting. The project aims to encourage maternal self-care while enabling fathers to participate actively in childcare and household responsibilities. It also provides an opportunity to integrate artificial intelligence, embedded hardware, physiological sensing, cloud-based data management and cross-platform mobile application development into a functional maternal wellness solution.

1.5 Objectives

The primary objective of MamaMind is to develop an integrated AI-powered maternal wellness and family support system that enables personalized self-care, passive health monitoring, baby-care management and collaborative parenting through a unified mobile application.

The specific objectives are:

- To develop dedicated Mother, Father and Baby dashboards using React Native and Expo.
- To integrate an ESP32-based wearable band for monitoring heart rate, SpO_2 and physical movement.
- To provide passive wellness monitoring and gentle recommendations based on sustained physiological changes.
- To implement an AI-powered voice companion using Google Gemini for

personalized assistance and parenting guidance.

- To support medication reminders, mood tracking, meditation, journaling and self-care activities.
- To enable baby-care tracking, vaccination reminders, milestone management and shared family responsibilities.
- To implement Firebase Authentication and Cloud Firestore for secure user management, activity storage and family synchronization.
- To evaluate application functionality, wearable data integration and AI-assisted wellness features through testing.

1.6 Scope and Limitations

The scope of MamaMind includes mobile application development, AI integration, wearable hardware implementation, cloud database management and maternal wellness monitoring. The application provides dedicated Mother, Father and Baby dashboards with features including voice assistance, medication reminders, guided meditation, mood journaling, relaxation games, shared tasks, feeding and sleep tracking, vaccination reminders and milestone management.

The wearable component incorporates an ESP32 microcontroller and sensors for monitoring heart rate, blood oxygen saturation and physical movement. Sensor readings are intended to support passive wellness awareness and context-sensitive recommendations. Google Gemini provides conversational assistance, while Firebase enables authentication, activity storage and family data synchronization.

The system has limitations related to sensor accuracy, connectivity, physiological variations and AI-generated recommendations. Wearable measurements may be affected by movement, sensor placement and environmental conditions. Elevated heart rate does not necessarily indicate stress, and the system cannot independently diagnose medical conditions. AI-generated suggestions provide general wellness support rather than professional medical advice. The application also depends on

appropriate internet connectivity for cloud synchronization and AI services.

1.7 Existing System

Existing maternal wellness solutions include pregnancy tracking applications, postpartum care platforms, medication reminder applications, wearable fitness trackers and baby-care management tools. Pregnancy and postpartum applications commonly provide educational information, developmental updates and wellness guidance, while baby-care applications support feeding, sleeping and diaper records.

Wearable devices provide physiological measurements such as heart rate, activity and blood oxygen saturation, but their integration with maternal-specific wellness recommendations may be limited. AI-based conversational applications offer general guidance and emotional support but may not incorporate wearable observations or shared parenting activities. These solutions provide useful individual capabilities but often lack the unified maternal wellness and family-support approach addressed by MamaMind.

1.8 Limitations of Existing Approaches

Conventional maternal wellness approaches face limitations related to fragmented functionality, manual tracking and limited integration between physiological monitoring and personalized assistance. Many applications focus on pregnancy information, baby-care records or general fitness independently, requiring users to switch between multiple platforms for different activities.

Manual health tracking can become inconvenient during demanding childcare routines, while general-purpose wearable applications may not provide maternal-specific wellness support. Family coordination tools may also lack

integration with maternal check-ins and baby-care activities. MamaMind addresses these limitations by combining AI assistance, passive wearable monitoring, personalized self-care and collaborative parenting within a connected mobile application.

1.9 Proposed MamaMind System

MamaMind consists of three interconnected components: a mobile application, an AI-powered assistance system and an ESP32-based wearable monitoring band. The mobile application provides dedicated Mother, Father and Baby dashboards developed using React Native, Expo and TypeScript. Firebase Authentication manages user access, while Cloud Firestore supports data storage, activity records and family synchronization.

The Mother dashboard provides wellness check-ins, voice-based AI assistance, medication reminders, meditation, journaling, relaxation activities and personalized recommendations. The Father dashboard supports shared responsibilities, encouragement messages, parenting guidance and bonding activities. The Baby dashboard manages feeding, sleep, diaper records, vaccination schedules, developmental milestones and memories.

The wearable band monitors heart rate, SpO_2 and physical movement, with sensor data intended to be transmitted to the application through Firebase integration. Sustained changes in physiological readings can be evaluated alongside activity context to provide gentle wellness suggestions. Google Gemini supports conversational assistance and personalized guidance. Together, these components establish an integrated platform for maternal self-care, wearable wellness awareness and collaborative family management.

1.10 Advantages and Applications

The primary advantage of MamaMind is the integration of maternal wellness assistance, wearable physiological monitoring, baby-care management and family collaboration within a single platform. Passive monitoring reduces the need for repeated manual observations, while AI-powered assistance provides accessible conversational support and personalized wellness suggestions. Dedicated parental dashboards encourage shared responsibilities and improve the organization of everyday caregiving activities.

MamaMind has potential applications in maternal wellness support, postpartum self-care, family caregiving, early childcare management and digital health education. Mothers can manage wellness routines, receive gentle reminders and access relaxation activities, while fathers can coordinate responsibilities and provide emotional support. The wearable system introduces opportunities for context-aware wellness assistance through physiological observations. Although MamaMind does not replace professional healthcare services or medical-grade monitoring equipment, it demonstrates a practical approach to combining artificial intelligence, wearable sensing and family-centered digital technologies to support maternal well-being and collaborative parenting.

LITERATURE SURVEY

2.1 Introduction

Maternal wellness and family support involve technologies for monitoring physiological conditions, managing emotional well-being, supporting childcare and improving communication between caregivers. Existing approaches include pregnancy tracking applications, wearable health monitoring devices, AI-powered conversational assistants, medication reminder systems and digital baby-care management platforms. Each approach offers different benefits depending on individual health needs, accessibility and available technological support.

This literature survey examines research in maternal healthcare, wearable physiological monitoring, artificial intelligence and family-centered digital health applications, along with existing maternal wellness platforms. The study provides a theoretical foundation for developing MamaMind, an integrated AI-powered maternal wellness and family support system with wearable health monitoring.

2.2 Review of Related Research

Maternal mental health research emphasizes the importance of emotional well-being and continuous support during pregnancy and postpartum recovery. Hahn-Holbrook et al. conducted a systematic review and meta-analysis of 291 studies across 56 countries to investigate postpartum depression prevalence and associated economic and health factors [1]. Upadhyay et al. examined postpartum depression in India through a systematic review and meta-analysis, highlighting its significance as a maternal public health concern [2].

Paulson and Bazemore investigated prenatal and postpartum depression among fathers and its association with maternal depression [3]. Their findings support the importance of considering both parents' emotional well-being. Cox et al. developed the Edinburgh Postnatal Depression Scale, a standardized screening instrument for identifying symptoms associated with postnatal depression [4]. These studies provide a foundation for MamaMind's maternal wellness check-ins, emotional support features and collaborative parenting approach.

Research on mobile-based maternal mental health interventions has demonstrated the potential of digital platforms to support emotional well-being. Tsai et al. evaluated the effectiveness and quality of mobile applications addressing perinatal depression and anxiety [5]. Miura et al. examined app-based interventions for preventing postpartum depression [6]. These studies provide background for MamaMind's personalized self-care recommendations, mood journaling and accessible wellness assistance.

Linardon et al. analyzed evidence from 176 randomized controlled trials investigating smartphone applications for depression and anxiety symptoms [7]. Another meta-analysis by Linardon et al. examined the effectiveness of mindfulness applications [8]. Fitzpatrick et al. evaluated Woebot, an automated conversational agent delivering cognitive behavioral therapy to young adults experiencing depression and anxiety symptoms [9]. These studies support the exploration of meditation, relaxation activities and conversational assistance in MamaMind, while recognizing that their findings do not directly establish the effectiveness of the proposed application.

2.3 Existing Maternal Wellness Technologies

Maternal wellness technologies include mobile applications for pregnancy tracking, emotional health monitoring, guided relaxation and postpartum support. Such applications commonly provide educational information, symptom records, reminders and self-care recommendations. Digital interventions can improve access to wellness resources, particularly when users face time constraints or limited opportunities for in-person support.

Studies by Tsai et al. and Miura et al. examined mobile interventions addressing perinatal depression, anxiety and postpartum mental health [5], [6]. Their findings provide a research basis for exploring accessible application-based support during pregnancy and postpartum recovery. However, intervention effectiveness depends on application design, user engagement and the quality of the support provided.

Wearable monitoring technologies provide an additional approach to observing physiological conditions through heart-rate, blood oxygen saturation and movement measurements. MamaMind incorporates an ESP32-based wearable band to support passive wellness awareness. The system is intended to observe sustained physiological changes and provide gentle recommendations while recognizing that sensor readings alone cannot establish emotional states or medical diagnoses.

2.4 AI-Based Wellness Assistance and Mindfulness Methods

Artificial intelligence enables digital applications to provide conversational assistance, personalized suggestions and context-aware responses. AI-powered systems can process natural language inputs and generate responses based on user requests. In maternal wellness applications, these capabilities can support emotional check-ins, relaxation guidance, parenting questions and

everyday self-care activities.

Fitzpatrick et al. investigated the effectiveness of an automated conversational agent for delivering cognitive behavioral therapy [9]. Their study demonstrates the potential of conversational interfaces in mental health support, although the results cannot be directly generalized to maternal healthcare. Linardon et al. also examined mindfulness-based applications and their effects on depression and anxiety symptoms [8], providing background for integrating guided meditation and relaxation activities.

Google Gemini introduces multimodal AI capabilities that can support natural language interaction and contextual assistance [11]. MamaMind integrates Gemini through an Express-based backend to provide an AI voice companion, wellness suggestions and parenting guidance. The application also incorporates meditation, journaling and relaxation activities to encourage regular self-care without presenting AI-generated responses as professional medical advice.

2.5 Existing Maternal Wellness and Support Systems

Existing maternal wellness solutions include mobile-based mental health interventions, mindfulness applications and AI-powered conversational support systems.

Perinatal Mental Health Applications: Mobile applications addressing perinatal depression and anxiety provide psychological support, educational resources, symptom tracking and self-guided activities. Tsai et al. evaluated their effectiveness and quality through a systematic review and meta-analysis [5]. Miura et al. further investigated app-based interventions for preventing postpartum depression [6]. These

studies demonstrate the relevance of mobile platforms for maternal mental health support.

Mindfulness and Mental Health Applications: Smartphone applications incorporating meditation, relaxation and mindfulness exercises aim to reduce psychological distress and encourage emotional well-being. Linardon et al. examined the effectiveness of mental health and mindfulness applications through randomized controlled trials [7], [8]. These findings provide background for MamaMind's guided meditation, mood journaling and relaxation features.

AI-Based Conversational Support: Woebot is an automated conversational agent investigated by Fitzpatrick et al. for delivering cognitive behavioral therapy to young adults [9]. The study demonstrates the feasibility of conversational interaction for structured psychological support. MamaMind extends the concept of conversational assistance into a maternal wellness context through AI-generated guidance and parenting-related interactions.

Table 2.1 Comparative Analysis of Existing Maternal Wellness Approaches

System	Primary Function	Key Features
Perinatal Mental Health Apps [5], [6]	Maternal-mental health support	Psychological guidance, symptom tracking, self-care interventions
Mindfulness Applications [8]	Emotional wellness and relaxation	Guided meditation, mindfulness exercises, stress-management activities
Woebot [9]	AI-based conversatal support	Automated conversation, structured cognitive behavioral therapy

2.6 Comparative Analysis

Existing maternal wellness technologies provide different capabilities depending on their intended applications. Perinatal mental health applications emphasize psychological support and symptom management, while mindfulness applications focus on relaxation and emotional well-being. Conversational agents demonstrate the potential of automated interaction for delivering structured support. However, these approaches do not necessarily integrate wearable physiological monitoring, maternal self-care and collaborative parenting within a single platform.

MamaMind combines selected capabilities through dedicated Mother, Father and Baby dashboards. The Mother dashboard provides AI-powered assistance, medication reminders, mood journaling and relaxation activities. The Father dashboard supports shared responsibilities, emotional encouragement and parenting guidance. The Baby dashboard manages feeding, sleep, vaccination and developmental milestone records.

The system uses React Native and Expo for mobile application development [12], [13], Firebase Cloud Firestore for synchronized data management [10], [14], and Google Gemini with an Express backend for conversational assistance [11], [15], [16]. The main contribution of MamaMind is the integration of these established technologies with an ESP32-based wearable monitoring prototype to provide a connected maternal wellness and family-support platform.

2.7 Research Gap

Existing research provides established approaches to postpartum mental health assessment, mobile wellness interventions, mindfulness applications

and AI-powered conversational support. Studies have examined maternal depression, paternal mental health, digital interventions and automated psychological assistance [1]–[9]. However, the reviewed approaches primarily address specific aspects of mental health or digital support rather than the complete combination of maternal wellness, passive physiological monitoring, baby-care management and shared family responsibilities.

MamaMind addresses this integration opportunity by developing a unified mobile application that combines AI-powered assistance, wearable heart-rate, SpO_2 and movement monitoring, personalized self-care activities, medication reminders, baby-care records and collaborative parenting. Dedicated dashboards support both parents while organizing caregiving activities and wellness information.

The project also explores passive wellness monitoring by considering sustained physiological changes and movement context before providing gentle recommendations. Its contribution lies in integrating artificial intelligence, embedded sensing and cloud-based application technologies into a family-centered maternal wellness prototype. Further testing and clinical validation would be necessary to establish physiological monitoring reliability, recommendation effectiveness and suitability for real-world maternal healthcare applications.

SYSTEM DESIGN

3.1 General

The primary requirement of MamaMind is to provide an integrated platform that supports maternal wellness, passive physiological monitoring and shared family responsibilities. The wearable hardware must acquire heart-rate, SpO_2 and movement measurements and transmit sensor information for application-level monitoring. The mobile application must provide personalized wellness assistance, medication reminders, meditation, journaling, baby-care tracking and family task management. The system must support AI-powered conversational interaction, secure user authentication and synchronized storage of relevant information. Firebase enables user-specific data management, while Google Gemini provides AI-generated responses through the application backend. These requirements are categorized into functional requirements, which define the operations performed by the system, and non-functional requirements, which address usability, responsiveness, security, reliability, accessibility and maintainability.

3.2 Requirement Analysis

The primary requirement of MamaMind is to provide an integrated platform that supports maternal wellness, passive physiological monitoring and shared family responsibilities. The wearable hardware must acquire heart-rate, SpO_2 and movement measurements and transmit sensor information for application-level monitoring. The mobile application must provide personalized wellness assistance, medication reminders, meditation, journaling, baby-care tracking and family task management. The system must support AI-powered conversational interaction, secure user authentication and synchronized storage of relevant information. Firebase enables user-specific data management, while the AI backend provides personalized responses and

wellness suggestions. These requirements are categorized into functional requirements, which define the operations performed by the system, and non-functional requirements, which address usability, responsiveness, security, reliability, accessibility and maintainability.

3.3 Functional and Non-Functional Requirements

The requirements of MamaMind are divided into functional and non-functional categories to ensure that the developed system provides the necessary capabilities while maintaining practical operating characteristics.

3.3.1 Functional Requirements

The wearable band must acquire physiological measurements, including heart rate, SpO_2 and physical movement, through sensors connected to the ESP32 microcontroller. The system must support transmission and storage of sensor readings for passive wellness monitoring. The Mother dashboard must provide AI-powered voice assistance, medication reminders, wellness check-ins, guided meditation, mood journaling, relaxation activities and personalized self-care suggestions. The Father dashboard must support shared tasks, encouragement messages, daily support recommendations and AI-assisted parenting guidance. The Baby dashboard must provide feeding, sleep and diaper tracking, vaccination reminders, milestone management and baby-care information. The application must support user registration, authentication, family connection and synchronized data management through Firebase. AI integration must enable conversational assistance and context-aware wellness recommendations.

3.3.2 Non-Functional Requirements

The non-functional requirements focus on usability, reliability, security,

responsiveness and maintainability. MamaMind must provide an intuitive mobile interface with understandable navigation, readable information and accessible features for parents managing daily responsibilities. Wearable measurements should be collected consistently under suitable operating conditions, although movement, sensor positioning and environmental factors may affect accuracy. Cloud synchronization must account for internet connectivity and authentication status, while user-specific information should be protected through appropriate Firebase security rules. AI-generated responses should remain supportive and avoid unsupported medical conclusions. The modular separation of wearable sensing, mobile interfaces, AI services and cloud data management should support testing, troubleshooting and future improvements. MamaMind is intended for general maternal wellness assistance rather than clinical diagnosis or emergency medical monitoring.

3.4 Overall System Architecture

MamaMind follows an integrated hardware–software architecture consisting of physiological data acquisition, cloud-based storage, AI processing and mobile application modules. The wearable band uses sensors connected to an ESP32 microcontroller to acquire heart-rate, SpO_2 and movement measurements. The ESP32 processes the available sensor readings and transmits them to Cloud Firestore through its configured network connection. The mobile application retrieves the stored information and presents it through the wearable health monitoring interface. Sustained physiological changes may be evaluated alongside movement information to generate appropriate wellness suggestions.

The mobile application is developed using React Native, Expo and TypeScript and provides separate Mother, Father and Baby dashboards. Firebase

Authentication manages user access, while Cloud Firestore stores activity records, family information and shared tasks. The AI backend processes conversational requests, parenting guidance and personalized wellness suggestions. This architecture separates hardware-based sensing from application interaction, AI processing and cloud data management while enabling the components to operate within a connected maternal wellness system.

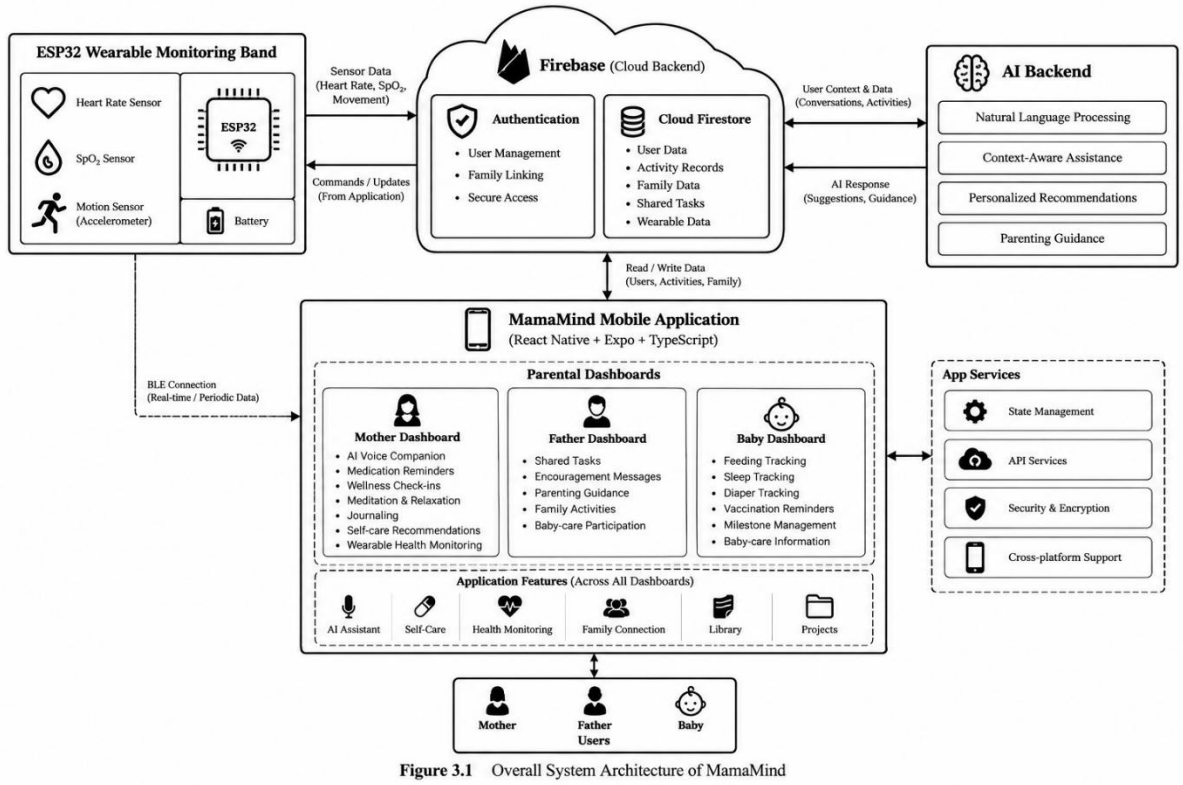


Figure 3.1 Overall System Architecture of MamaMind

Figure 3.1 Overall system architecture of MamaMind

3.5 Hardware–Software Integration

Hardware–software integration enables physiological measurements acquired by the wearable band to be used within the MamaMind application. The ESP32 coordinates sensor acquisition, basic processing and data transmission. Heart-rate, SpO_2 and movement information are transferred to Firebase, where

the application can retrieve and display the latest available readings. This arrangement enables the wearable hardware and mobile software to operate as interconnected components without requiring continuous manual entry of physiological observations.

The passive wellness monitoring workflow demonstrates the integration between hardware and software. When the wearable detects sustained elevated heart-rate readings, the monitoring logic can consider movement information and measurement reliability before generating a gentle wellness suggestion. For example, the application may recommend taking a short break or practicing relaxation when appropriate. Such recommendations are intended to encourage self-care rather than identify medical conditions. The integration also supports future improvements involving personalized monitoring rules, historical trends and more advanced sensor-data analysis.

3.6 System Flow Diagram

The MamaMind workflow begins when a user registers or logs into the application and selects the appropriate parental role. Firebase Authentication verifies the user, after which the corresponding Mother or Father dashboard is displayed. The Mother dashboard provides access to AI assistance, wellness check-ins, medication reminders, meditation, journaling, community features and baby-care activities. The Father dashboard supports family tasks, encouragement, parenting guidance and baby-care participation. The Baby dashboard provides feeding, sleep, diaper, vaccination and milestone management.

In parallel, the ESP32-based wearable band collects heart-rate, SpO_2 and movement readings. The sensor information is transmitted to Firebase and retrieved by the application for monitoring. When sustained physiological changes are identified, the system may provide context-aware wellness

suggestions. User activities and shared family information are stored in Firestore, allowing relevant information to remain available across connected dashboards. AI-related requests are processed through the backend before the generated responses are returned to the application.

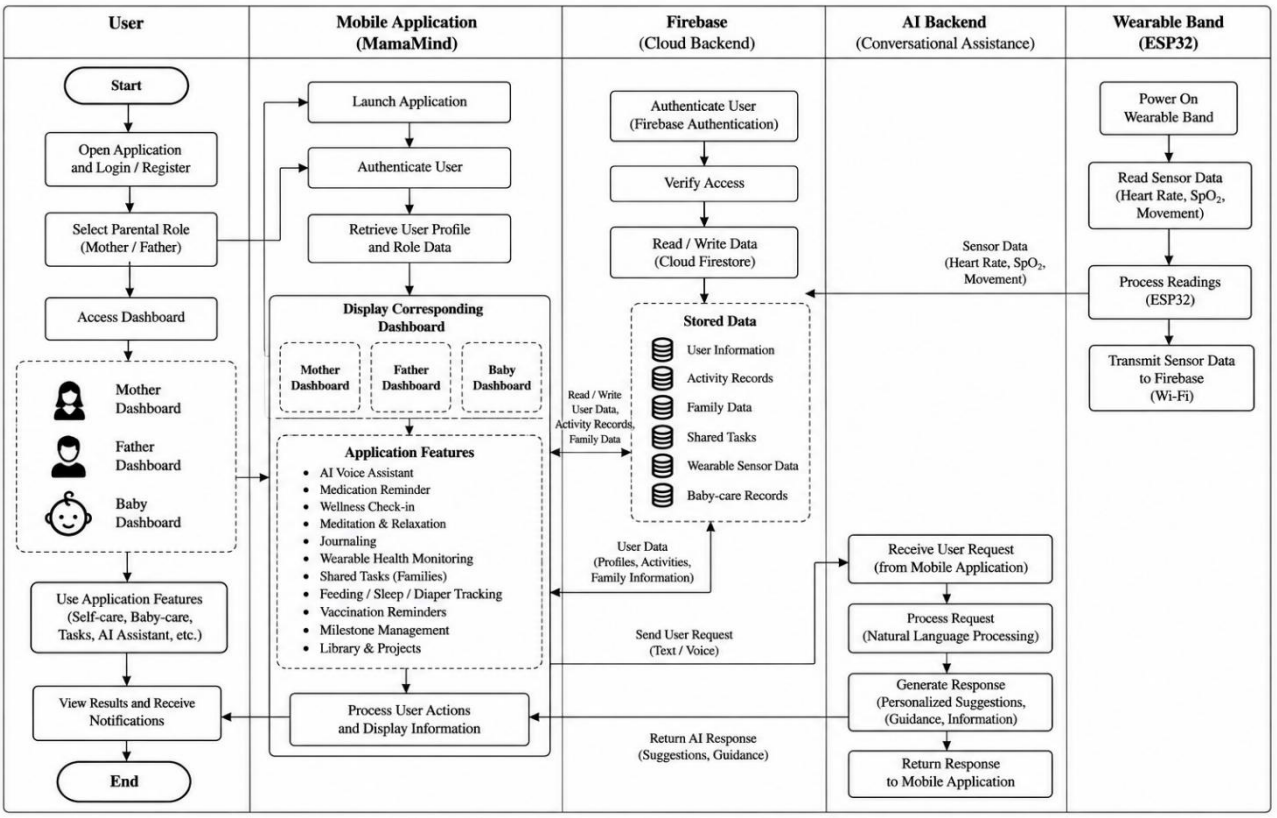


Figure 3.2 Overall system flow diagram of MamaMind

Figure 3.2 Overall system flow diagram

3.7 Use Case Diagram

The use case diagram illustrates the interactions between the Mother, Father and Baby-care functions within the MamaMind system. The Mother can access AI voice assistance, manage medication reminders, perform wellness check-ins, participate in meditation, maintain journal entries, use relaxation activities and monitor wearable health information. The Father can view daily support suggestions, send

encouragement messages, manage shared responsibilities, receive AI-assisted parenting guidance and participate in baby-care activities. The Baby dashboard enables authorized parents to record feeding, sleep and diaper activities, manage vaccination information and track developmental milestones. Firebase Authentication manages user access, while Cloud Firestore synchronizes family information and shared tasks. The ESP32-based wearable band functions as an external monitoring component that supplies physiological readings to the application. The diagram represents the relationships between users and the system's primary functions, demonstrating how MamaMind supports maternal wellness and collaborative parenting.

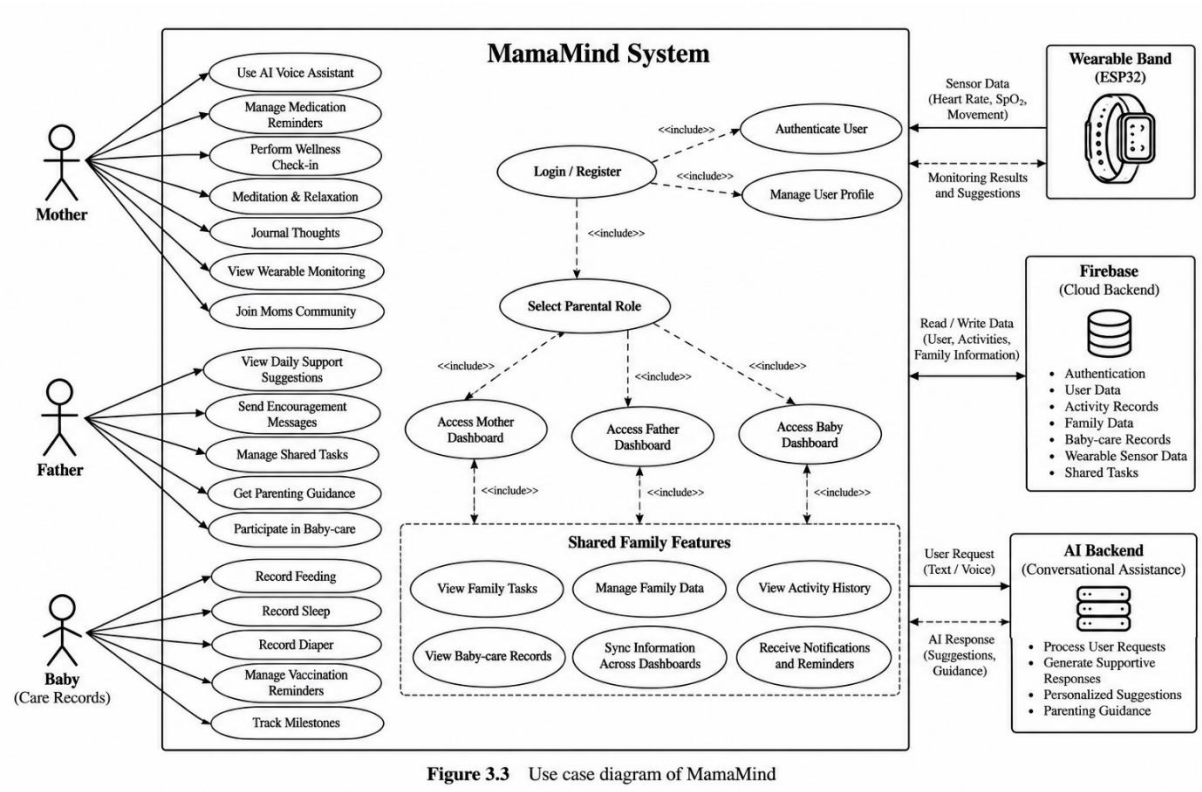


Figure 3.3 Use case diagram of MamaMind

Figure 3.3 Use case diagram of MamaMind

3.8 Firebase Communication and Data Flow

Firebase provides the cloud-based communication and data management infrastructure for MamaMind by integrating user authentication, persistent storage and family synchronization. Firebase Authentication manages registration and login, while Cloud Firestore stores user profiles, wellness activities, medication records, baby-care information, shared tasks and wearable sensor measurements. The ESP32-based wearable band transmits heart-rate, SpO_2 and movement readings to Firebase, where the mobile application retrieves the latest available information for passive wellness monitoring. Mother and Father dashboards access authorized shared records, enabling coordinated family responsibilities and baby-care management. AI-powered requests are processed through the application backend, while relevant user activities and wellness information can be stored for personalized assistance. This architecture supports organized data exchange, persistent records and synchronized interaction between wearable hardware, cloud services and the MamaMind application.

3.9 Module Design

MamaMind is organized into hardware, application, AI and cloud-service modules according to their responsibilities. The wearable sensing module uses the ESP32 microcontroller and connected sensors to acquire heart-rate, SpO_2 and movement information. The physiological monitoring module supports processing and interpretation of available readings for passive wellness awareness. The Firebase integration module manages authentication, user information, activity records and synchronized family data. These modules collectively support communication between wearable hardware and the mobile application.

The software modules include the Mother Dashboard, Father Dashboard, Baby Dashboard, AI Companion, Self-Care, Medication Management, Family

Connection and Wearable Health Monitoring. The Mother Dashboard manages wellness activities, reminders, journaling and relaxation features. The Father Dashboard provides parenting assistance, encouragement and shared responsibilities, while the Baby Dashboard supports feeding, sleep, vaccination and milestone records. The AI Companion uses the backend for conversational assistance, and the Family Connection module manages shared caregiving activities through Firestore. The modular design supports independent testing, maintenance and future enhancements without unnecessarily affecting unrelated application functions.

HARDWARE IMPLEMENTATION

4.1 Wearable Health Monitoring Overview

The wearable health monitoring band is the hardware component of the MamaMind system, developed to collect physiological and movement-related information and support passive maternal wellness monitoring. The hardware is built around an ESP32 microcontroller integrated with sensors for measuring heart rate, blood oxygen saturation (SpO_2) and physical movement. The ESP32 receives sensor measurements, performs basic processing and transmits the collected information to Firebase Cloud Firestore through Wi-Fi connectivity. The MamaMind mobile application retrieves these readings and displays them through its health monitoring interface. This arrangement enables physiological observations to be collected without requiring mothers to manually enter health measurements. The wearable system is intended to support wellness awareness and context-sensitive self-care recommendations during pregnancy and postpartum recovery.

The wearable monitoring system focuses on three primary parameters: heart rate, SpO_2 and movement. Heart-rate measurements provide information about cardiovascular activity, while SpO_2 readings estimate peripheral blood oxygen saturation. Movement measurements help identify physical activity and provide additional context when interpreting physiological changes. When elevated heart-rate readings persist, the proposed monitoring logic considers movement information before generating gentle wellness suggestions, such as encouraging rest or relaxation. The system is designed for general wellness assistance rather than clinical diagnosis or emergency medical monitoring.

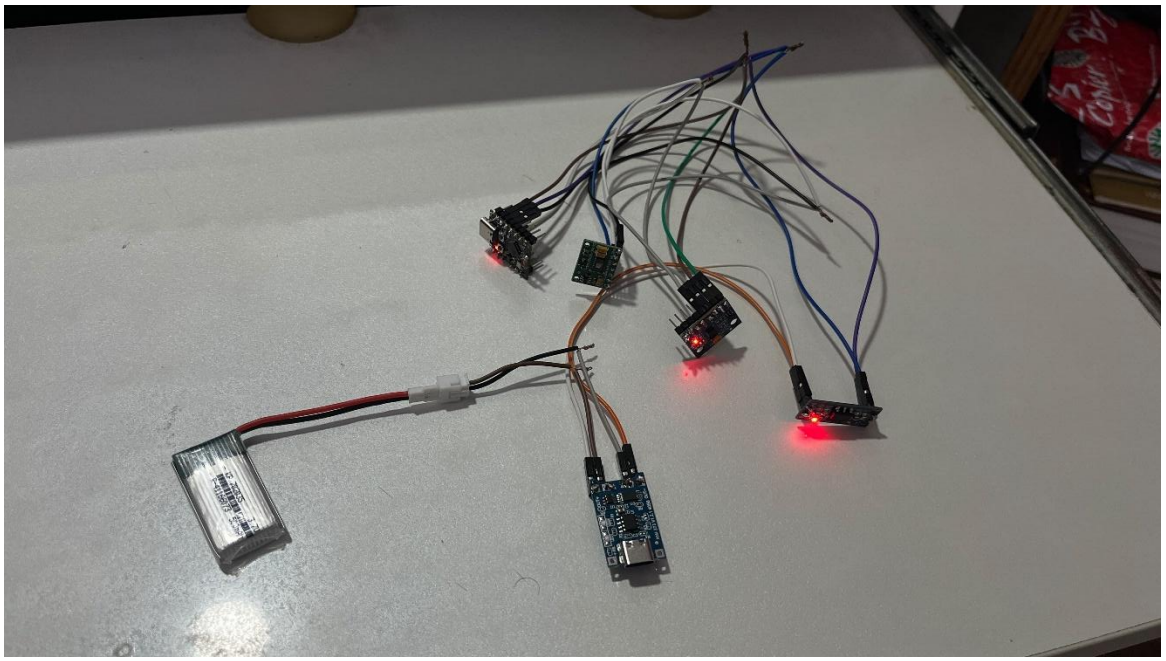


Figure 4.1 Working hardware prototype

4.2 ESP32 Microcontroller

The ESP32 serves as the central processing and communication unit of the MamaMind wearable monitoring system. It provides General-Purpose Input/Output (GPIO) pins, digital communication interfaces and built-in Wi-Fi connectivity, enabling communication between the connected physiological sensors and Firebase. The microcontroller receives heart-rate, SpO_2 and movement measurements, processes the available readings and prepares the data for wireless transmission. Its programmable architecture supports sensor acquisition, basic filtering, data formatting and communication management. The ESP32 is suitable for the wearable prototype because of its compact design, processing capabilities and support for wireless communication.

4.3 Heart Rate Monitoring Sensor Sensor

The heart-rate monitoring sensor measures pulse-related physiological changes to estimate the user's heart rate in beats per minute (BPM). Wearable heart-rate monitoring commonly uses photoplethysmography (PPG), an optical technique that detects changes in blood volume through variations in reflected or transmitted light. The sensor provides measurement information to the ESP32 for processing and transmission to the MamaMind application. The collected readings are used to display heart-rate information and support the identification of sustained changes over time. However, measurement quality can be influenced by movement, skin contact, sensor positioning and environmental conditions. The system therefore treats heart-rate measurements as wellness indicators rather than direct evidence of emotional stress or medical abnormalities.

4.4 Blood Oxygen Saturation (SpO_2) Monitoring

The Blood oxygen saturation monitoring provides an estimate of the percentage of oxygen-saturated hemoglobin in peripheral blood. Optical pulse-oximetry sensors commonly use red and infrared light measurements to estimate SpO_2 through differences in light absorption. In the MamaMind wearable system, the SpO_2 sensing component is intended to provide oxygen saturation measurements that can be processed by the ESP32 and displayed within the mobile application. These readings contribute to the overall physiological monitoring interface and support awareness of available health information. SpO_2 measurements may be affected by motion, sensor contact, circulation and device limitations. Consequently, the wearable is not intended to replace clinically validated pulse oximeters or professional medical assessment.

4.5 Movement Detection Sensor

The movement detection component is designed to identify physical activity

and provide contextual information for interpreting wearable physiological readings. Motion sensing can be implemented using an accelerometer, which measures acceleration along multiple axes and detects changes associated with movement. The collected signals are processed by the ESP32 to determine whether movement has occurred and to update the corresponding information within the MamaMind application. Movement data is particularly useful when evaluating heart-rate variations because physical activity can naturally increase heart rate. By considering movement alongside physiological measurements, the system aims to reduce unnecessary relaxation suggestions during ordinary activity. The movement information supports general activity awareness rather than precise clinical movement analysis.

4.6 Wireless Communication and Firebase Integration

Wireless communication enables the MamaMind wearable band to transfer physiological measurements to the mobile application through cloud-based synchronization. The ESP32 uses Wi-Fi connectivity to transmit available heart-rate, SpO_2 and movement readings to Firebase Cloud Firestore. The stored measurements are retrieved by the MamaMind application and displayed within the wearable health monitoring interface. This communication structure separates hardware-based sensor acquisition from mobile application processing and presentation. The system depends on network availability for transmitting new measurements, while the application can distinguish recently received readings from older stored information. Firebase integration also provides a foundation for maintaining monitoring history and supporting future improvements in personalized wellness assistance.

4.7 Circuit Connections and Hardware Assembly

The MamaMind wearable monitoring circuit consists of the ESP32 microcontroller and connected heart-rate, SpO_2 and movement sensing components. The physiological sensors communicate with the ESP32 through their supported digital interfaces, while the movement sensing component provides activity-related measurements. The microcontroller coordinates sensor acquisition, processes the received information and transmits formatted readings through Wi-Fi. All connected components require compatible supply voltages and a common electrical ground to support reliable operation. The hardware assembly is designed to accommodate the sensing components within a wearable arrangement while maintaining suitable sensor positioning and contact with the user. The exact wiring configuration depends on the selected sensor modules and the ESP32 development board used in the prototype.

Table 4.1 Hardware Components and Their Functions

Component	**Function**	**Communication**
ESP32 Microcontroller	*Sensor processing and wireless data transmission*	*Wi-Fi*
Heart Rate Sensor	*Measures pulse rate in BPM*	*Sensor interface to ESP32*
SpO_2 Sensor	*Estimates blood oxygen saturation*	*Sensor interface to ESP32*
Movement Sensor	*Detects physical movement*	*Sensor interface to ESP32*
Firebase Cloud Firestore	*Stores and synchronizes sensor readings*	*Internet connection*

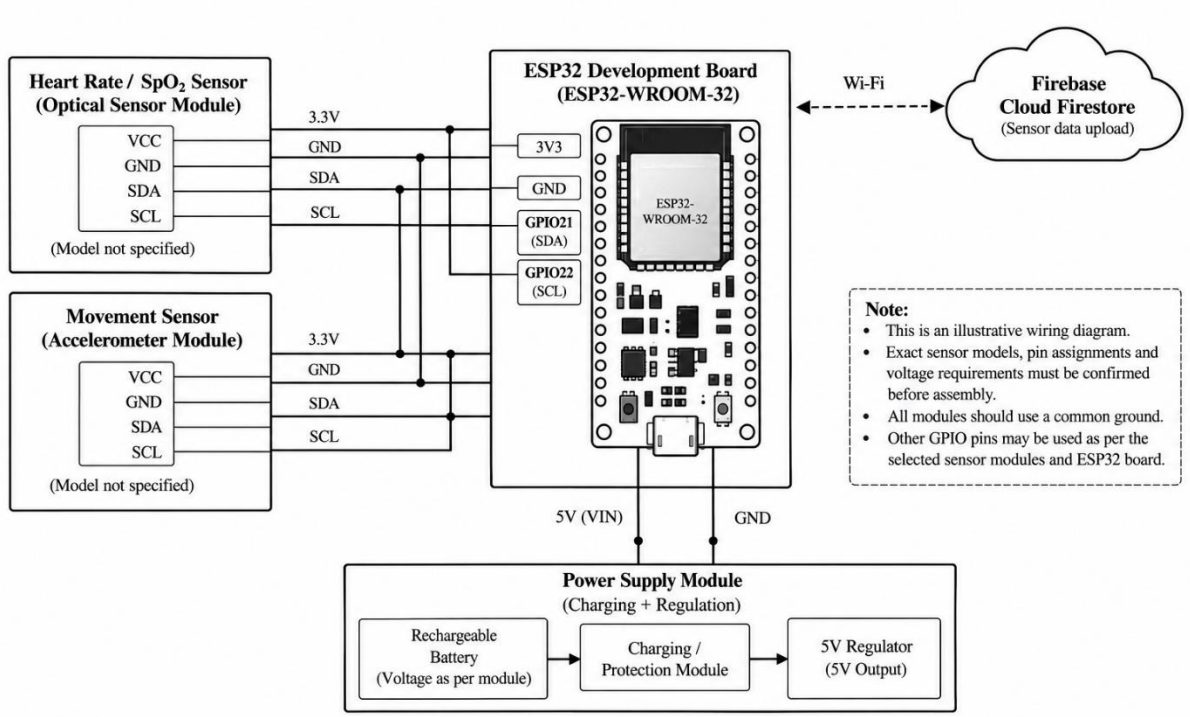


Figure 4.2 MamaMind wearable hardware connection diagram

Figure 4.2 MamaMind wearable hardware connection diagram

4.8 Working Hardware Prototype

The MamaMind wearable hardware prototype integrates physiological sensing, embedded processing and wireless data transmission to support passive maternal wellness monitoring. The ESP32 receives available heart-rate, SpO_2 and movement measurements from the connected sensors and transmits the processed information to Firebase Cloud Firestore. The MamaMind mobile application retrieves these readings and presents them through the health monitoring interface, enabling users to observe physiological information alongside other maternal wellness features. The proposed monitoring approach also considers sustained heart-rate changes and movement context to provide gentle self-care suggestions when appropriate.

The prototype demonstrates the intended connection between embedded wearable sensing and cloud-based maternal wellness assistance. Hardware

evaluation should examine sensor responsiveness, measurement consistency, wireless transmission and synchronization with the mobile application. Detailed testing procedures and observed results are presented in Chapter 7, based on the functionality verified during implementation. The wearable component provides a foundation for further development of context-aware monitoring and personalized wellness support within MamaMind.

SOFTWARE IMPLEMENTATION

5.1 Software Architecture and Technology Stack

The software architecture of MamaMind integrates mobile application development, cloud-based data management, artificial intelligence and wearable health monitoring within a unified maternal wellness support system. The application consists of three primary functional interfaces: the Mother Dashboard, Father Dashboard and Baby Dashboard. These interfaces provide personalized access to maternal self-care, emotional support, shared parenting responsibilities and infant care management. The system is developed using React Native with Expo and TypeScript, enabling a mobile-oriented interface with reusable components and organized navigation. Firebase Authentication manages user access, while Cloud Firestore supports persistent storage and synchronization of application data. An Express-based backend handles AI-related requests, allowing the mobile application to provide conversational assistance and personalized wellness suggestions.

The software follows a modular architecture in which authentication, maternal wellness, baby care, family collaboration, AI assistance and wearable monitoring operate as interconnected components. Expo Router manages navigation between application screens, while React hooks support component state and user interactions. Firebase services maintain information such as user profiles, daily care activities, medication records, baby-care logs and shared family tasks. The ESP32 wearable component transmits available physiological readings to Cloud Firestore through Wi-Fi, allowing the mobile application to retrieve and display monitoring information. This separation of responsibilities improves software organization, supports data consistency and provides a foundation for future expansion.

5.2 User Authentication and Role-Based Navigation

MamaMind incorporates user authentication and role-based navigation to provide appropriate application features for mothers and fathers. During registration, users provide the required account information and select their corresponding role. Firebase Authentication manages account creation and sign-in, while Cloud Firestore stores associated user profiles and application-specific details. After successful authentication, the application directs users to the appropriate dashboard based on their selected role. This approach allows both parents to access relevant features while maintaining a consistent application structure.

The system also supports family connectivity through an invitation-code mechanism that associates participating parents with a shared family record. Once connected, mothers and fathers can coordinate selected baby-care responsibilities, access shared tasks and exchange supportive information. Role-based interfaces help organize the available functions without requiring users to navigate through unrelated features. Authentication and Firestore access controls provide the foundation for protecting personal information and restricting access to authorized records.

5.3 MamaMind Mobile Application

MamaMind is a mobile application designed to support maternal wellness, infant care and shared parenting through a user-friendly digital environment. Its interface uses a warm colour palette, clearly organized cards and simplified navigation to reduce interaction complexity for new parents. The application provides access to personalized wellness activities, medication reminders, baby-care records, family collaboration and AI-assisted support. The home interfaces are organized according to parental roles, allowing users to locate

frequently required functions without navigating through numerous unrelated screens.

The application combines manual interactions with automated assistance to reduce repetitive user input. Users can record daily activities, maintain baby-care information, receive reminders and access supportive recommendations through the same interface. Firebase synchronization helps maintain relevant information across connected accounts, while the AI backend supports conversational responses and context-sensitive guidance. This integration establishes MamaMind as a centralized platform for maternal self-care and family-oriented baby-care management.

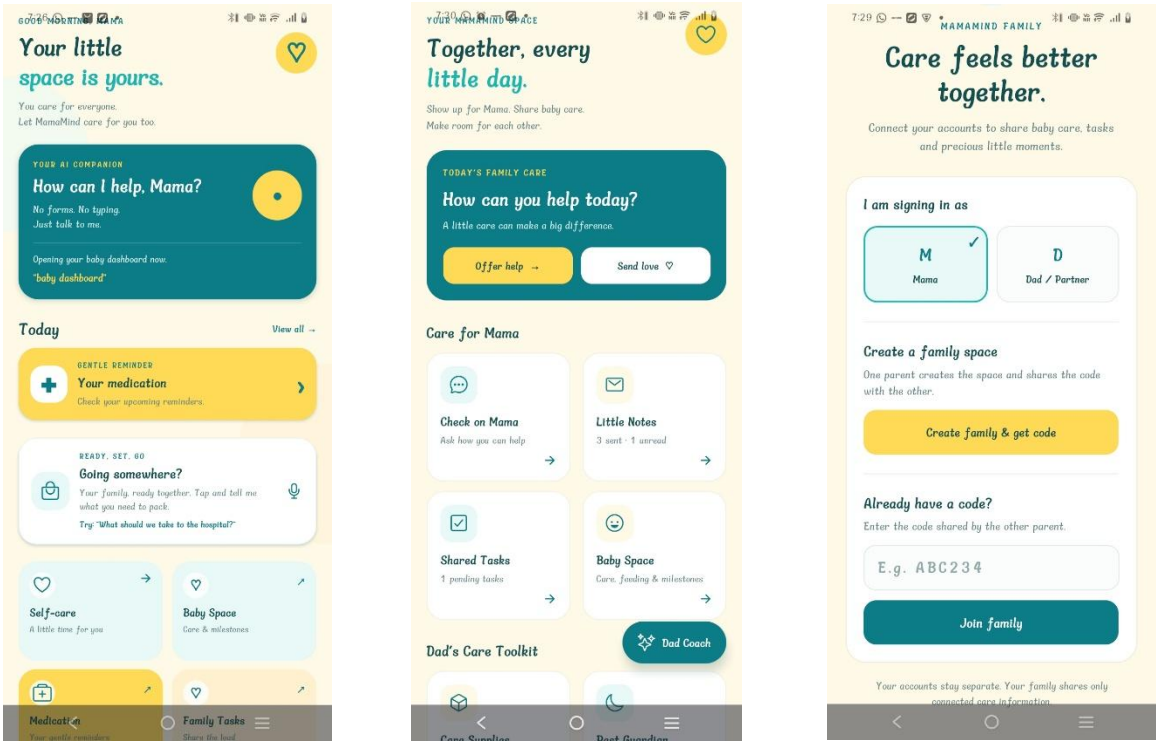


Figure 5.1 Home screen of MamaMind Application

5.4 Mother Dashboard and Personalized Wellness

The Mother Dashboard serves as the primary interface for maternal self-care and emotional wellness support. It provides access to daily care activities, guided meditation, medication management, AI companionship, community interaction and baby-care shortcuts. The dashboard presents frequently used functions through clearly labelled cards and supportive messages, allowing mothers to select activities according to their needs. The interface is designed to encourage manageable wellness routines rather than impose rigid schedules or require extensive daily questionnaires.

Personalized wellness support is developed around user interactions, recorded activities and available check-in information. The application can use these inputs to provide gentle reminders and relevant self-care suggestions, including rest, relaxation and short mindfulness activities. The dashboard also provides access to family collaboration features, enabling mothers to share responsibilities with their partners. By combining wellness activities and parenting assistance within a single interface, the Mother Dashboard supports a more balanced approach to postpartum care.

5.5 Father Dashboard and Shared Parenting Support

The Father Dashboard is designed to encourage active paternal involvement in maternal wellness and infant care. It provides access to shared parenting tasks, baby-care information, supportive suggestions and family-related activities. The dashboard includes features that help fathers understand their partner's shared wellness check-ins, offer encouragement and participate in daily responsibilities. This approach recognizes the importance of emotional and practical support from both parents during the postpartum period.

The application incorporates shared task management to coordinate responsibilities such as baby-care activities and household support. Connected parents can view relevant tasks and update their completion status through Firebase synchronization. The Father Dashboard also includes an AI-assisted coaching feature that provides general

parenting guidance, bonding suggestions and supportive communication ideas. These functions aim to promote mutual understanding and distribute caregiving responsibilities more effectively.

5.6 Baby Dashboard and Infant Care Management

The Baby Dashboard provides a centralized interface for maintaining essential infant-care information. It supports baby profile details, including date of birth and relevant care notes, along with records of feeding, sleep and diaper changes. Parents can use the dashboard to organize routine care activities and review previously recorded information. The interface also includes features for tracking vaccinations, developmental milestones, baby supplies and memorable moments, helping families maintain important information in an accessible format.

The dashboard organizes infant-care functions into separate sections to simplify data entry and retrieval. Feeding and sleep records support routine tracking, while vaccination reminders assist parents in managing upcoming care requirements. Milestone and memory features allow families to document developmental experiences through notes and photographs. Age-related information, including food guidance, is presented as general reference material rather than individualized medical advice. By integrating these functions, the Baby Dashboard reduces the need for multiple separate applications or manual records.

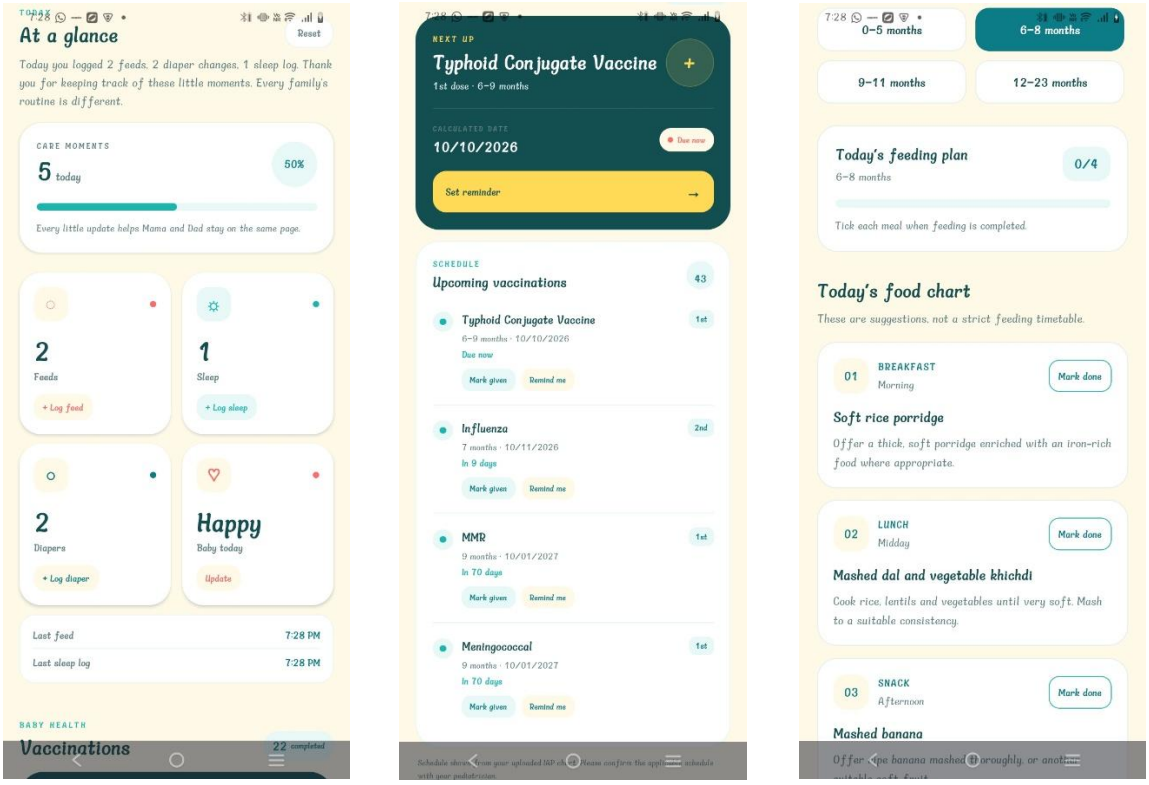


Figure 5.2 Image of Baby Dashboard And Baby Care

5.7 Self-Care, Medication Reminders and Wellness Activities

MamaMind incorporates self-care features designed to encourage consistent maternal wellness practices through simple and manageable activities. The application provides guided meditation sessions, daily self-care tasks, journaling and a Happy Jar feature for recording positive experiences. It also includes interactive relaxation activities such as Bubble Pop, Fidget Spinner, Pebble Stack, Zen Sand, Mood Garden and Sound Garden. These activities offer optional opportunities for relaxation and emotional expression without requiring prolonged interaction.

The medication management feature allows users to maintain medicine details, configure reminder times and record whether scheduled medication has been taken. Local notifications support reminder delivery, while stored records help maintain continuity between application sessions. Activity

logging can capture relevant interactions, including meditation sessions, completed self-care tasks and medication updates. These records provide context for future personalized reminders and wellness suggestions. The features are intended to support routine organization and self-care engagement rather than replace professional medical advice.

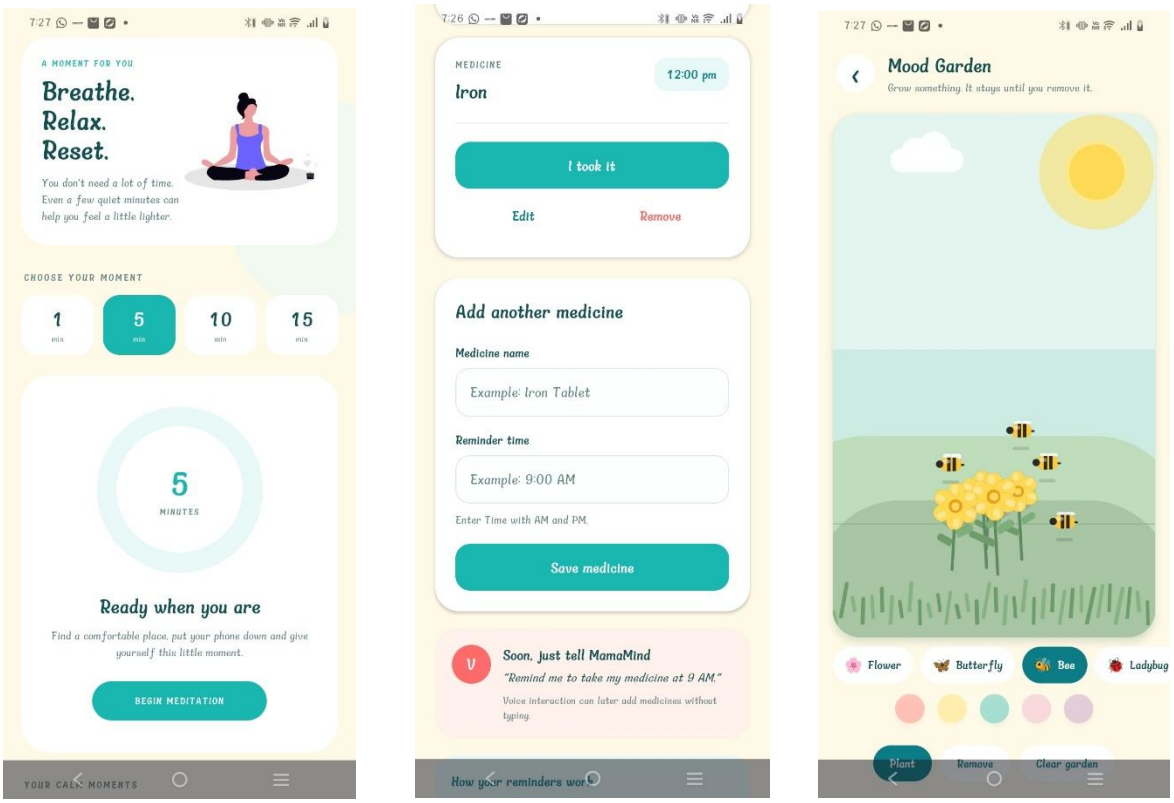


Figure 5.3 Image of self care and wellness activities

5.8 AI Companion and Personalized Assistance

The MamaMind AI Companion provides conversational assistance for maternal wellness, parenting concerns and everyday emotional support. Users can interact with the assistant through text-based conversations and supported voice interactions. The mobile application sends relevant requests to an Express-based backend, which processes the conversation and returns generated responses to the interface. This separation allows the mobile application to manage user interaction while the backend coordinates AI-related processing.

The assistant is designed to provide supportive responses, general wellness information and practical suggestions based on the context shared by users. Related features include personalized self-care guidance, the Father Dashboard's coaching assistant and MamaGo, which helps prepare baby-related outing checklists. The application may use relevant activity information and user-provided context to improve the usefulness of its recommendations. However, the AI Companion is not intended to diagnose medical conditions, determine a user's emotional state with clinical certainty or replace qualified healthcare professionals.

5.9 Firebase Integration and Wearable Health Monitoring

Firebase provides the primary cloud-based data management infrastructure for MamaMind. Firebase Authentication supports secure account access, while Cloud Firestore maintains application information such as user profiles, family associations, medications, daily care activities, shared tasks and baby-care records. Firestore listeners allow supported interfaces to reflect updated information without requiring users to manually reload the application. This synchronization is particularly useful for shared parenting features, where both parents need access to current task and care information.

The wearable health monitoring component extends the application by connecting physiological measurements with the digital wellness interface. The ESP32-based wearable transmits available heart-rate, SpO_2 and movement information to Cloud Firestore through Wi-Fi. The MamaMind application retrieves these measurements and displays them within the health monitoring screen. The intended monitoring logic considers sustained physiological changes alongside movement information to support context-sensitive wellness suggestions. The system distinguishes general wellness observations from medical conclusions and does not treat wearable measurements as a substitute for clinical monitoring.

METHODOLOGY AND ALGORITHMS

6.1 Overall System Methodology

MamaMind follows an integrated methodology that combines maternal wellness management, infant-care tracking, family collaboration, AI-powered assistance and wearable health monitoring. The system begins with user authentication and role identification, after which mothers and fathers access their respective dashboards. Users can record daily activities, manage medication reminders, maintain baby-care information and interact with the AI companion. Firebase Authentication manages user access, while Cloud Firestore stores and synchronizes application data. The ESP32-based wearable monitoring component provides physiological measurements that can be accessed through the mobile application. These interconnected processes support personalized wellness assistance and shared parenting responsibilities within a unified digital environment.

The methodology is organized into five primary stages: user authentication, activity and care-data collection, cloud synchronization, context-based processing and presentation of relevant information or recommendations. Each stage operates through a separate software module while sharing necessary information through the application and cloud infrastructure. This modular approach supports maintainability and allows wellness features to function independently when certain inputs, such as wearable measurements, are unavailable.

6.2 User Authentication and Role Identification

The authentication methodology verifies user credentials and identifies the

appropriate application role before providing access to the system. During registration, Firebase Authentication creates an account, while the corresponding user profile and selected role are stored in Cloud Firestore. Following successful login, the application retrieves the user's role and directs the session to the Mother Dashboard or Father Dashboard. Role-based navigation ensures that users receive features appropriate to their responsibilities while allowing authorized family information to be shared between connected accounts.

The system also supports family association through an invitation-code mechanism. A parent can share the generated family code with their partner, enabling the application to associate both accounts with a common family identifier. This identifier is used to organize shared tasks and selected family-related records. Access to private information must remain subject to authenticated-user permissions and appropriate Firestore security rules.
Role identification:

$$D(u) = \begin{cases} D_m, & \text{if } R(u) = \text{Mother} \\ D_f, & \text{if } R(u) = \text{Father} \end{cases}$$

Where $D(u)$ represents the dashboard assigned to user u, $R(u)$represents the user's registered role, D_m is the Mother Dashboard and D_f is the Father Dashboard

6.3 Daily Activity Logging and Wellness Data Processing

MamaMind uses activity logging to maintain records of selected maternal wellness and baby-care interactions. Activities such as meditation, medication intake, self-care tasks and baby-care events are recorded with relevant information, including the activity type, completion status and timestamp. These records provide a structured representation of user interactions and can be used to display activity history or

support personalized reminders. The application stores supported records through Firebase, allowing information to remain available across sessions.

The recorded information can also be organized into daily summaries to identify completed and pending activities. For example, medication records indicate whether a scheduled medicine has been marked as taken, while meditation history maintains information about completed sessions. These summaries support wellness awareness without assigning clinical interpretations to ordinary application behaviour.

Daily activity completion percentage:

$$C_d = \frac{N_c}{N_t} \times 100$$

Where C_d represents the daily completion percentage, N_c is the number of completed activities and N_t is the total number of scheduled activities for the day, provided $N_t > 0$

This calculation is a general method for summarizing task completion and does not represent a medical or psychological wellness score.

6.4 Medication Reminder and Notification Methodology

The medication reminder module allows users to enter medicine details, specify scheduled times and maintain medication intake records. When a reminder is configured, the application stores the corresponding medication information and schedules a local notification using Expo Notifications. At the scheduled time, the notification prompts the user to review the medicine reminder. Users can subsequently mark the medication as taken, allowing the

application to update the associated record and display its current status.

The reminder methodology separates notification delivery from medication confirmation. A notification being displayed does not automatically indicate that the medicine has been consumed. Instead, the application relies on explicit user confirmation to update the intake status. This distinction improves record clarity and avoids treating scheduled reminders as completed medication events.

Medication status:

$$M_i = \begin{cases} 1, & \text{if medication } i \text{ is confirmed as taken} \\ 0, & \text{otherwise} \end{cases}$$

Medication adherence based on recorded confirmations:

$$A_m = \frac{\sum_{i=1}^{N} M_i}{N} \times 100$$

Where A_m represents the percentage of scheduled medication events confirmed as taken and N represents the total number of scheduled medication events in the selected period. This measure reflects user-recorded confirmations and does not independently verify medication consumption.

6.5 Wearable Sensor Data Acquisition

The wearable monitoring methodology collects heart-rate, blood oxygen saturation (SpO_2) and movement information through sensors connected to the ESP32 microcontroller. The microcontroller receives available sensor readings, prepares the measurements for transmission and sends the information to Cloud Firestore through Wi-Fi connectivity. The MamaMind mobile application retrieves the stored measurements and presents them through its health monitoring interface. Each reading can be associated with a timestamp to distinguish recently collected measurements from older observations.

Sensor data acquisition is intended to support passive wellness awareness rather than continuous clinical supervision. Heart-rate measurements are represented in beats per minute, SpO_2 values are expressed as percentages, and movement information provides context regarding physical activity. Sensor contact, movement artifacts, communication interruptions and hardware limitations can affect the reliability or availability of measurements. Consequently, the system should identify missing or outdated data rather than substitute randomly generated readings for actual wearable measurements.

Sensor reading representation:

$$S_t = \left[HR_t,\ SpO_{2,t},\ MV_t,\ T_t\right]$$

Where HR_t represents heart rate, $SpO_{2,t}$ represents oxygen saturation, MV_t represents movement information and T_t represents the measurement timestamp.

6.6 Physiological Data Processing and Trend Analysis

Physiological data processing aims to identify sustained changes in wearable measurements while considering their relationship with physical activity. Individual sensor readings may fluctuate because of movement, sensor contact or temporary physiological changes. A moving-average technique can be used to reduce short-term variations before evaluating trends. The processed heart-rate measurements can then be compared with a configurable reference level to identify periods of sustained elevation. Movement information provides additional context because physical activity can naturally increase heart rate.

The proposed trend-analysis method avoids interpreting a single elevated reading as evidence of stress or a medical condition. Instead, it evaluates multiple valid measurements collected over a selected observation window. If elevated readings persist while movement remains relatively low, the application may offer a gentle

suggestion to rest or begin a relaxation activity. This approach represents a proposed wellness-support algorithm; its thresholds and effectiveness require implementation testing and validation before being described as verified functionality.

Moving-average heart rate:

$$\overline{HR}_t = \frac{1}{N}\sum_{i=0}^{N-1} H\,R_{t-i}$$

Where $\overline{HR}_t$ is the average heart rate over the selected observation window and N is the number of valid measurements.

Sustained elevation condition:

$$E_t = \begin{cases} 1, & \text{if } \overline{HR}_t > H_{\text{ref}} \text{ for duration } D \\ 0, & \text{otherwise} \end{cases}$$

Where H_{ref} is a configurable reference value and D is the required observation duration. These parameters must be selected and validated appropriately; they are not established medical thresholds.

6.7 Context-Aware Wellness Recommendation Methodology

The context-aware wellness recommendation methodology combines available user activity information, self-reported wellness inputs and wearable measurements to support personalized self-care suggestions. The application can consider recent meditation activity, daily care completion, medication reminders and relevant physiological trends when determining which wellness information to present. Recommendations are intended to be supportive and non-diagnostic, encouraging manageable actions such as taking a short break, practising guided meditation or reviewing pending self-care activities.

The recommendation process follows a condition-based approach in which

available context determines the type of suggestion. For example, when sustained heart-rate elevation is observed alongside low movement, the system may suggest a relaxation activity. When no reliable wearable data is available, recommendations can instead be based on user-selected activities and recorded preferences. The system should avoid interpreting physiological changes as definite indicators of emotional distress and should not delay medical assistance when users report concerning symptoms.
Illustrative recommendation condition:

$$R_t = \begin{cases} R_{\text{relax}}, & E_t = 1 \text{ and } MV_t < M_{\text{ref}} \\ R_{\text{routine}}, & \text{if a scheduled self-care task is pending} \\ R_{\text{general}}, & \text{otherwise} \end{cases}$$

Where R_t represents the selected recommendation category, R_{relax} represents a gentle relaxation suggestion, R_{routine} represents a routine reminder and R_{general} represents general wellness guidance. The conditions illustrate the intended methodology rather than a clinically validated decision model.

6.8 AI-Powered Conversational Assistance

The AI-powered conversational methodology enables users to obtain supportive responses through text and supported voice interactions. The user submits a question, concern or request through the MamaMind interface, and the application forwards the relevant information to its backend. The backend processes the request using an AI service and returns a generated response, which is displayed in the conversation interface. For voice-based interactions, speech recognition converts spoken input into text, while text-to-speech functionality can present the response audibly.

The AI Companion supports general maternal wellness guidance, emotional encouragement, parenting information and self-care suggestions. Additional

AI-assisted functions include father-oriented parenting support and MamaGo outing checklist generation. The quality and relevance of generated responses depend on the user's request and the information made available to the backend. The assistant is therefore positioned as a supportive information tool rather than a clinical decision-making system. Medical concerns, emergencies and potentially serious symptoms require appropriate professional evaluation.

6.9 Shared Parenting and Baby-Care Data Synchronization

The shared parenting methodology allows connected family members to coordinate selected responsibilities through Firebase Cloud Firestore. After two parental accounts are associated with a common family identifier, shared task records can be created, retrieved and updated by authorized users. Each task contains information such as its description and completion status, with additional metadata maintained where supported. Firestore synchronization allows connected interfaces to reflect changes made by either parent, helping reduce duplicate work and improve communication.

Baby-care information is managed through structured records for feeding, sleep, diaper changes, vaccinations and developmental milestones. The application stores the relevant details and presents them through the Baby Dashboard for later review. These records provide an organized history of care activities and support routine planning. The system does not independently verify the accuracy of manually entered information, and vaccination schedules or age-related guidance should be confirmed against appropriate healthcare recommendations.

Shared task completion percentage:

$$P_s = \frac{T_c}{T_a} \times 100$$

Where P_s represents the percentage of shared tasks completed, T_c is the number of completed tasks and T_a is the total number of assigned tasks, provided $T_a > 0$.

6.10 Data Storage and Firebase Synchronization

MamaMind uses Firebase Cloud Firestore to maintain structured application records and synchronize information between supported modules. User profiles, medication information, daily care activities, meditation history, journal entries, family tasks and baby-care records are organized into appropriate collections or documents. The application performs database operations according to the requirements of each module, including creating new records, retrieving saved information and updating completion states. Firestore listeners support the retrieval of changes as they occur, allowing the interface to display updated information.

Wearable monitoring data follows a separate acquisition path in which the ESP32 transmits available measurements to Firestore through Wi-Fi. The mobile application retrieves these measurements for display and further wellness-oriented processing. Authentication and database security rules are necessary to restrict access to authorized information. The system also requires appropriate handling of unavailable network connections, outdated sensor readings and incomplete records to maintain reliable application behaviour.

This chapter retains the methodology-and-algorithms style of your reference, including mathematical expressions where relevant to MamaMind. The equations for completion percentages, moving averages and context-based recommendations describe methods that can be implemented or evaluated; they should not be presented as already tested unless they match your actual code.

TESTING AND RESULTS

7.1 Testing Strategy

Testing and evaluation of MamaMind focus on verifying the functionality, reliability and integration of the developed maternal wellness and family support system. The evaluation covers user authentication, role-based navigation, maternal self-care features, medication reminders, baby-care management, shared parenting tasks, AI-powered assistance and wearable health monitoring. Functional testing examines whether individual modules respond correctly to user interactions, store relevant information and display appropriate outputs. Integration testing focuses on communication between the React Native application, Firebase services, AI backend and ESP32 wearable monitoring component. The testing strategy also considers data persistence, notification behaviour, synchronization between connected accounts and handling of unavailable sensor measurements. These procedures provide a structured basis for assessing the system's operation, although comprehensive clinical validation and long-term usability evaluation remain outside the present project scope.

7.2 Experimental Setup

The MamaMind testing environment consists of the React Native mobile application developed using Expo and TypeScript, Firebase Authentication, Cloud Firestore and an Express-based backend for AI-related functions. The application can be executed on a compatible Android device for evaluating navigation, user interactions, notifications and database operations. Firebase services provide the environment for verifying account creation, login, data storage and synchronization between connected users. The AI backend supports testing of conversational requests and generated responses, while the wearable monitoring setup includes an ESP32 microcontroller and sensors

intended to measure heart rate, SpO_2 and movement.

Testing is organized around representative user activities, including registration, dashboard navigation, medication scheduling, meditation completion, baby-care recording and shared task updates. The wearable monitoring interface is evaluated separately to distinguish actual sensor measurements from simulated or unavailable data. The experimental setup supports functional verification of the software modules and their communication interfaces without assuming that all hardware features have undergone clinical or laboratory validation..

7.3 User Authentication and Dashboard Testing

Authentication testing examines the registration, login and role-based navigation functions of MamaMind. The test procedure involves creating user accounts, entering valid and invalid credentials and verifying the application's response to authentication requests. Following successful login, the application retrieves the registered user role and directs the session to the corresponding Mother or Father Dashboard. The testing also considers navigation between screens, account-related error messages and the handling of incomplete user information.

Role-based testing verifies that mothers and fathers are presented with their respective features. The Mother Dashboard provides access to self-care, medication, meditation, AI companionship and baby-care functions, while the Father Dashboard emphasizes shared responsibilities, parenting guidance and family support. These tests establish whether the application's navigation structure corresponds to the intended role-based design.

7.4 Maternal Self-Care and Medication Reminder Testing

The maternal self-care modules are evaluated through representative interactions

with meditation, daily care activities, journaling and relaxation features. Testing includes opening individual screens, completing supported activities and verifying whether the corresponding information is reflected in the application. Meditation testing examines session initiation, timer behaviour and completion recording, while daily care testing focuses on task creation, status updates and persistence of saved information. Journaling and Happy Jar features are examined for their ability to maintain and retrieve user-generated entries.

Medication reminder testing focuses on medicine entry, scheduled notification configuration and intake-status management. Users enter medication details and select reminder times, after which the application schedules the corresponding notification. The testing procedure examines whether saved medication information remains available and whether marking a medicine as taken updates its recorded status. Notification delivery must be verified on the target Android device because operating-system permissions, battery restrictions and notification settings can affect reminder behaviour.

7.5 Baby-Care Management Testing

Baby-care testing evaluates the functions provided through the Baby Dashboard, including profile management, feeding records, sleep tracking, diaper logs, vaccination information and developmental milestones. The procedure involves entering representative baby-care details, saving the records and retrieving them through the relevant interface. Testing examines whether information is displayed in the appropriate section and whether updates are reflected after navigating away from and returning to the screen.

The baby profile module is evaluated for maintaining details such as date of birth and care notes, while routine tracking modules are examined for their ability to organize recorded activities. Vaccination and milestone features are

tested for information entry, display and reminder behaviour where implemented. These tests focus on application functionality and data organization rather than assessing the medical accuracy of age-based guidance or developmental interpretations.

further calibration refinement was required to obtain closer agreement with the selected digital reference values. These results established the baseline for evaluating subsequent calibration improvements rather than providing evidence of certified colour-measurement accuracy.

7.6 Shared Parenting and Firebase Synchronization Testing

Shared parenting testing examines the connection between Mother and Father accounts and the synchronization of selected family information. The procedure involves associating two accounts through the family invitation mechanism and creating shared tasks from one account. The connected account is then used to retrieve the task information and update its completion status. Testing evaluates whether changes are stored correctly in Cloud Firestore and reflected in the relevant application interfaces.

Firebase integration is also examined through selected data operations, including saving user profiles, updating daily care records and retrieving previously stored information. These tests assess whether application state and cloud records remain consistent during ordinary interactions. Additional scenarios include incomplete records, network interruptions and attempts to access information without appropriate authorization. The evaluation establishes the intended behaviour of shared family features, while broader security and concurrency testing would be required for production deployment.

7.7 AI Companion and Voice Interaction Testing

The AI Companion is evaluated using representative requests related to maternal

wellness, self-care, parenting support and general emotional encouragement. The testing procedure involves submitting text queries through the application, forwarding them to the backend and examining whether a response is returned and displayed. Additional scenarios can include incomplete requests, unavailable backend connectivity and questions requiring referral to professional medical assistance. The evaluation focuses on request handling, response presentation and the appropriateness of general supportive guidance rather than measuring clinical effectiveness.

Voice interaction testing examines supported speech recognition and text-to-speech functions. Spoken requests are provided through the application interface, and the recognized text is checked for its ability to initiate the intended interaction. Voice navigation is evaluated using supported screen-related commands, while response playback is examined for basic operation. Recognition performance may vary with pronunciation, background noise, microphone permissions and device capabilities. The AI assistant is not considered a substitute for professional healthcare advice.

7.8 Wearable Health Monitoring Testing

Wearable monitoring testing focuses on the acquisition, transmission and presentation of heart-rate, SpO_2 and movement information. The intended procedure involves obtaining measurements from sensors connected to the ESP32, transmitting the readings through Wi-Fi to Cloud Firestore and retrieving them through the MamaMind mobile application. Testing examines whether the application displays the received values with appropriate timestamps and whether updated measurements replace older displayed information.

The monitoring interface must distinguish actual hardware readings from simulated values used during software development. Where live sensor integration has not been verified, testing results should be reported as interface

or data-flow demonstrations rather than completed physiological measurement experiments. Additional testing is required to evaluate sensor accuracy, transmission delays, measurement consistency and behaviour during network interruptions. The proposed sustained-trend monitoring logic also requires validation before its recommendations can be described as reliable indicators of physiological or emotional conditions.

7.9 Functional and Integration Testing

Functional and integration testing provides a structured method for examining the interaction between MamaMind's major software modules. The evaluation considers account authentication, role-based navigation, self-care activities, medication reminders, baby-care management, family synchronization, AI interactions and wearable data presentation. Each function is assessed using representative input conditions and expected application behaviour. The results should be recorded as passed, failed or pending according to the actual observations obtained during testing.

7.10 Limitations and Discussion

The testing methodology demonstrates how MamaMind can be evaluated as an integrated maternal wellness and family support application. The proposed test cases cover the principal software functions and their interactions with authentication, cloud storage, AI assistance and wearable monitoring. However, functional demonstrations alone do not establish long-term reliability, medical effectiveness or clinical accuracy. The application relies on user-entered information for several wellness and baby-care functions, and the completeness of these records depends on consistent user participation. Notification delivery may vary across Android devices, while cloud-based features require suitable network connectivity.

Wearable monitoring introduces additional limitations related to sensor positioning,

measurement quality, data transmission and the interpretation of physiological changes. Heart-rate and SpO_2 readings should not be treated as clinically validated without comparison against appropriate reference instruments. Similarly, AI-generated responses and context-aware wellness suggestions require further evaluation for safety, consistency and relevance. Future testing should include repeated hardware measurements, multi-device compatibility checks, structured usability assessments and longer-term observation of application performance. These improvements would provide stronger evidence of the system's reliability and practical usefulness.

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

MamaMind: AI-Powered Maternal Wellness and Family Support System with Wearable Health Monitoring demonstrates the development of an integrated mobile application designed to support maternal well-being, infant care and shared parenting responsibilities. The system combines React Native, Expo, Firebase and AI-powered assistance to provide a unified platform for managing daily self-care activities, medication reminders, emotional support and baby-care information. The Mother Dashboard offers personalized wellness features, including meditation, journaling, relaxation activities and AI companionship, while the Father Dashboard encourages active participation through shared tasks, parenting guidance and supportive interactions. The Baby Dashboard provides organized tracking of feeding, sleep, vaccinations and developmental milestones. The proposed ESP32-based wearable monitoring component extends the system by enabling heart-rate, SpO_2 and movement measurements to be transmitted through Wi-Fi and accessed within the mobile application. By integrating these functions, MamaMind establishes a foundation for accessible, family-centered maternal wellness assistance. The system emphasizes preventive awareness, emotional encouragement and coordinated care rather than medical diagnosis or clinical treatment.

8.2 Project Achievements

The major achievement of MamaMind is the integration of maternal self-care, infant-care management and family collaboration within a single mobile application. The system provides dedicated Mother and Father Dashboards with role-based navigation, allowing users to access functions appropriate to their responsibilities. Firebase Authentication supports account management, while Cloud Firestore enables

persistent data storage and synchronization of shared parenting information. The application incorporates medication reminders, guided meditation, daily care activities, journaling, relaxation games and a supportive mothers' community. The Baby Dashboard organizes essential infant-care records, including feeding, sleep, diaper changes, vaccination information and developmental milestones. AI-powered conversational assistance provides general wellness guidance, emotional encouragement and parenting support through text and supported voice interactions. The project also establishes a software framework for integrating ESP32-based physiological monitoring with the mobile application through cloud communication. Together, these features demonstrate the practical application of mobile development, cloud computing, AI integration and embedded sensing in a maternal wellness support system

8.3 Current Limitations

Although MamaMind provides an integrated approach to maternal wellness and family support, several limitations remain. The accuracy and completeness of wellness records depend on information entered by users, while cloud-based synchronization and AI assistance require reliable internet connectivity. Medication notifications may be affected by device permissions, operating-system restrictions and background execution policies. AI-generated responses are intended for general information and emotional encouragement and cannot replace professional medical advice or psychological assessment. Similarly, wearable measurements may be influenced by sensor positioning, movement artifacts, hardware characteristics and communication interruptions. Heart-rate and SpO_2 readings require validation against appropriate reference instruments before they can be considered clinically reliable. The proposed context-aware monitoring approach also requires further evaluation to determine suitable observation periods and recommendation conditions. Comprehensive usability studies, long-term performance testing and clinical validation have not been established within the present project scope. Therefore, MamaMind should be regarded as a supportive wellness application rather than a diagnostic or emergency

medical monitoring system.

8.4 Future Enhancements

Future development of MamaMind can focus on improving wearable hardware integration, physiological monitoring reliability and personalized wellness assistance. A compact wearable design with optimized power consumption and improved sensor placement could support more convenient daily use. Further refinement of ESP32 firmware, Wi-Fi communication and Firebase synchronization could improve the consistency of sensor-data transmission. Additional evaluation using appropriate reference devices would help assess heart-rate and SpO_2 measurement accuracy, while validated trend-analysis methods could improve the relevance of context-aware wellness suggestions. Offline data buffering and automatic synchronization could also reduce the impact of temporary network interruptions.

The mobile application could be enhanced through more advanced personalization based on user preferences, daily routines and recorded activities. Improved AI-assisted features could provide multilingual conversational support, personalized self-care planning and more accessible voice-based interaction. The Father Dashboard could be expanded with additional family coordination tools, parenting resources and shared scheduling functions. The Baby Dashboard could incorporate customizable vaccination reminders, improved growth visualization and age-appropriate developmental information supported by verified healthcare references. These enhancements would improve usability while maintaining the application's non-diagnostic purpose.

Future evaluation could include structured usability studies involving mothers and fathers, accessibility testing across different Android devices and extended observation of application reliability. Stronger privacy controls,

secure data-sharing mechanisms and transparent management of sensitive family information would further support responsible deployment. Collaboration with maternal healthcare professionals could help review educational content, improve safety guidance and assess the practical relevance of wellness recommendations. These developments would build upon the existing MamaMind application and support its evolution into a more comprehensive, accessible and reliable maternal wellness and family support platform.

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